Root system is classified into two types.

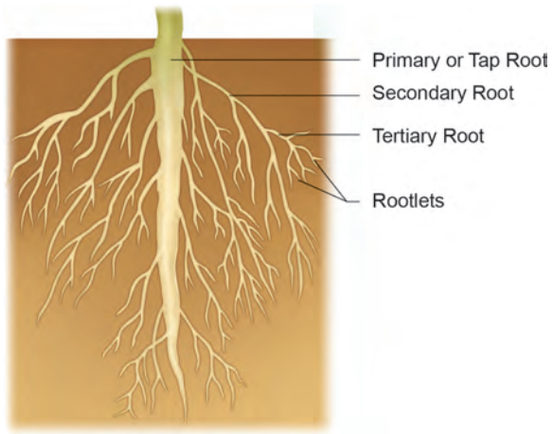
a. Taproot system

b. Fibrous root system

a. Taproot system

It consists of a single root, called **taproot**, which grows straight down into the ground. Smaller roots, called lateral roots arise from the taproot. They are seen in dicotyledonous plants.

Example: Bean, Mango, Neem.



b. Fibrous root system

It consists of a cluster of roots arising from the base of the stem. They are thin and uniform in size. It is generally seen in monocotyledonous plants.

Example: Grass, Paddy, Maize.



Fibrous Root of Grass

Activity 1

Water absorption by Root

Aim: To observe absorption of water by root.

What you need? A carrot, a glass of water and blue ink.

What to do? Place a carrot in a glass of water with a few drops of blue ink. Leave the carrot in water for two to three days. Then cut the carrot into half length wise and observe.

What do you learn? Blue colour appears in carrot which indicates the upward movement of water in the carrot showing that root conducts water.

Functions of the Root

- Fixes the plant to the soil.
- Absorbs water and minerals from the soil.
- Some plants like carrot and beet root store food in root.

Guess it



Is it a root or stem?

2. Shoot system

The aerial part of the plant body above the ground is known as the **shoot system**. Main axis of the shoot system is called the **stem**. The shoot system consists of stem, leaves, flowers and fruits.

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